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320413 (20)

BE (4th Semester)

Examination, April-May, 2011

Branch : Civil

SURVEYING II

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Answer questions (Part A and Part B) from all 5 units. Part (A) of all units is compulsory

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(2 marks). Answer any 2 questions out of 3 (2×7=14) from Part B. Assume suitable missing data if any.

Unit - I

Part - 'A'

1. (a) Define an eccentric station in triangulation. 2

Part - 'B'

(b) Explain the phenomenon of "phase of the signal" 7

(i) when signal is partly in shade.

(c) During the triangulation survey of a hilly part of the country, two triangulation stations P

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and Q are selected, distance PQ being 120 km. Elevations above MSL are P = 210 m, Q = 1050 m. Intervening grounds L and M have elevations 330 m and 557 m. PL = 50 km, PM = 80 km. Determine if P and Q are intervisible, and if necessary, find the minimum height of scaffolding at Q, assuming P as ground station. Line of sights is to clear peaks by 3 m. 7

B and C are the stations in a triangulation survey. A satellite station

S is set up near C such that S is due south of C. Given AC = 7246.4 m, BC = 6756.1 m, CS = 44.5 m, ∠ASC = 62° 35′ 20″, ∠ASB = 72° 58′ 30″. Calculate angle ACB. 7

Unit - II

Part - A

Q. 2. (a) Define what is Station Adjustment and Network Adjustment in triangulation ? 2

Part - B

(b) Determine the Most Probable Values of angles A, B and C taken at a triangulation station 7

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Observation	Weight
A = 72° 12′ 42.3″	3
B = 53° 18′ 53.6″	4
C = 110° 24′ 48.5″	2
A + B = 125° 31′ 36.8″	2
B + C = 163° 43′ 44.6″	2
A + B + C = 235° 56′ 26.2″	1

(c) The following angles were measured at station O to close the horizon.

∠AOB = 83° 42′ 28.75″ weight 3.

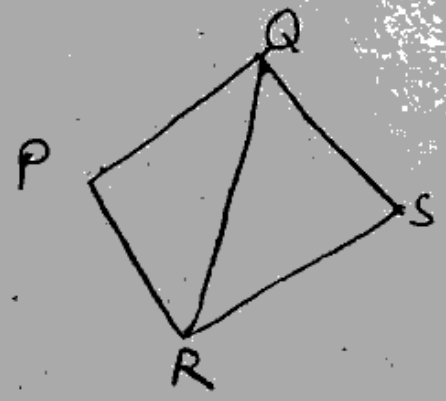
∠BOC = 102° 15′ 43.26″ weight 2

∠COD = 94° 38′ 27.22″ weight 4

∠DOA = 79° 23′ 23.77″ weight 2

Adjust the angles by the method of correlates. 7

(d) Figure below shows two connected triangles PQR and QRS. Adjust the angles. 7



∠P = 68° 12′ 33″

∠Q = 128° 16′ 30″

∠R = 111° 02′ 30″

∠S = 52° 28′ 35″

$Q_1 = 62° \ 18' \ 30''$

$Q_2 = 65° \ 57' \ 55''$

$R_1 = 49° \ 29' \ 00''$

$R_2 = 61° \ 33' \ 35''$

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Unit - III

Part - A

3. (a) Explain the principal use of tacheometry. 2

Part - B

(b) A line was levelled with a tacheometer fitted with an anallatic lens. Following observations are made with staff vertical. Compute elevations of A, B and C. Instrument was fitted with an anallatic lens. 7

Inst. Stn	Ht of Axis	Staff at	Vertical Angle	Staff Readings			Remarks
				Bottom	Centre	Top	
A	1.38	BM	−1° 54′	1.020	1.720	2.420	RL = 638.55
A	1.38	B	+2° 36′	1.220	1.825	2.430	
B	1.40	C	+3° 06′	0.785	1.610	2.435	

(c) Derive expression for vertical elevations and horizontal distances for fixed hair tacheometric method, Inclined Sight – Elevation, Staff held Vertical. 7

(d) A tacheometer is placed at a station A and readings on a staff held on BM of RL 100.00 and a station B are 0.640, 2.200, 3.760 and 0.0010, 2.120, 4.230 respectively. The

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angles of depression are 6° 19′ and 7° 42′. Find the distance AB and RL of B. The constants are 100 and 0.3. 7

Unit - IV

Part - A

Q. 4. (a) Define "Crab" and "Drift" effects in an aerial photograph. 2

Part - B

(b) What is a tilt distortion ? Prove that in a tilted photograph, tilt distortion is radial from the isocentre. 7

(c) With a neat figure explain the method to determine scale of an aerial photograph. What do you mean by datum scale and average scale ? 7

(d) With a neat sketch explain the method of computations of length of line between points of different elevations from measurements on a vertical photograph. 7

Unit - V

Part – A

(a) Define echosounding. Name the instrument used to perform the same. 2

Part – B

(b) Describe the graphical methods of solving 3 point problem in hydrographic surveying. 7

(c) In order to locate the position of O of a boat, observations were made with a sextant to three points A, B and C on the shore. The angles AOB and BOC were found to be 50° 56′ and 27° 30′. From the plan, AB and BC are 450 m and 220 m while angle ABC was 163° 18′. What are the distances of O from A, B and C ? 7

(d) Describe the following methods of locatio; f Soundings 7

(i) Range and time intervals

(ii) Range and two angles from shore

(iii) Range and two angles from boat

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